



SKYLENA INFO TECHNOLOGY PVT. LTD.

Innovation of Thought

Corporate Training | Software Development | Cloud Services | Cyber Security Services

CIN : U62099TZ2023PTC029167

Project Description

Students will learn the fundamentals of **Defensive Cyber Security (Blue Team)** by monitoring systems, detecting threats, analyzing logs, hardening operating systems, configuring security controls, and responding to simulated security incidents in a controlled lab environment.

Day 1 – Introduction to Defensive Cyber Security & Lab Setup

Explanation

Defensive Cyber Security focuses on protecting systems, networks, and data from cyber threats. Students will understand the role of a Security Operations Center (SOC) and set up a security lab.

Tools

- Windows 11
- Kali Linux
- VirtualBox / VMware
- GitHub

Task

- Set up the virtual lab.
- Configure Windows and Kali Linux virtual machines.

Day 2 – Operating System Security

Explanation

Operating system security involves securing user accounts, permissions, services, updates, and configurations to minimize security risks.

Tools

- Windows Security
- Windows Defender
- Local Security Policy
- Linux Terminal

Task

- Configure user accounts.
- Review security settings and permissions.

Day 3 – Network Security Fundamentals

Explanation

Network security protects systems from unauthorized access by understanding IP addressing, ports, protocols, firewalls, and secure communication.

Tools

- Wireshark
- ipconfig
- ping
- tracert
- netstat

Task

- Capture network information.
- Identify active connections and network devices.

Day 4 – Firewall Configuration

Explanation

Firewalls filter incoming and outgoing traffic based on security rules, helping prevent unauthorized access.

Tools

- Windows Defender Firewall
- UFW (Ubuntu Firewall)

Task

- Create inbound and outbound firewall rules.
- Block and allow selected services.

Day 5 – Log Management

Explanation

System logs record events and activities. Security analysts review logs to detect suspicious behavior and investigate incidents.

Tools

- Windows Event Viewer
- Linux Syslog
- Event Log Explorer

Task

- Analyze login events and security logs.

Day 6 – Security Monitoring

Explanation

Security monitoring continuously observes systems and networks to identify suspicious activities and security threats.

Tools

- Wazuh
- OSSEC
- Windows Event Viewer

Task

- Monitor system events and identify suspicious activities.

Day 7 – Vulnerability Management

Explanation

Vulnerability management identifies, prioritizes, and remediates security weaknesses before attackers can exploit them.

Tools

- Nessus Essentials
- OpenVAS
- Microsoft Defender Vulnerability Management

Task

- Perform a vulnerability scan.
- Prepare a remediation report.

Day 8 – Malware Analysis Basics

Explanation

Malware analysis helps identify malicious files and understand their behavior without executing them on production systems.

Tools

- VirusTotal
- Hybrid Analysis
- Any.Run (Community Edition)

Task

- Analyze sample files safely.
- Document Indicators of Compromise (IOCs).

Day 9 – Email Security & Phishing Detection

Explanation

Phishing attacks attempt to steal credentials or distribute malware through deceptive emails. Students learn how to identify phishing indicators.

Tools

- VirusTotal
- MXToolbox
- Google Safe Browsing

Task

- Analyze phishing email samples.
- Identify suspicious URLs and attachments.

Day 10 – Endpoint Security

Explanation

Endpoint security protects desktops, laptops, and servers using antivirus, endpoint detection, encryption, and device management.

Tools

- Microsoft Defender Antivirus
- Windows Security
- BitLocker

Task

- Configure endpoint protection settings.

Day 11 – Incident Response

Explanation

Incident response is a structured process for detecting, containing, eradicating, and recovering from cyber security incidents.

Tools

- TheHive
- Cortex
- Microsoft Word

Task

- Develop an Incident Response Plan for a ransomware scenario.

Day 12 – Security Information & Event Management (SIEM)

Explanation

SIEM platforms collect, correlate, and analyze logs from multiple systems to detect security threats in real time.

Tools

- Splunk Free
- Wazuh Dashboard
- Elastic (ELK Stack)

Task

- Import logs and create a dashboard to monitor security events.

Day 13 – Security Policies & Compliance

Explanation

Organizations implement security policies and compliance standards to protect data and meet regulatory requirements.

Tools

- CIS Benchmarks
- ISO 27001 Documentation
- Microsoft Office

Task

- Create a Password Policy and Acceptable Use Policy.

Day 14 – Security Hardening

Explanation

Security hardening reduces vulnerabilities by disabling unnecessary services, applying updates, enforcing strong configurations, and securing applications.

Tools

- CIS Benchmarks
- Microsoft Security Compliance Toolkit
- Windows Defender Firewall

Task

- Harden a Windows system using security best practices.

Day 15 – Final Security Assessment & Presentation

Explanation

Students will demonstrate their security monitoring, vulnerability assessment, incident response, and system hardening activities through a final presentation.

Tools

- Microsoft PowerPoint
- GitHub
- Microsoft Word

Task

- Present the final project.
- Submit documentation and security reports.

Capstone Project

Enterprise Security Monitoring & Incident Response

Modules

- Security Lab Setup
- Operating System Hardening
- Network Monitoring
- Firewall Configuration
- Log Analysis
- Security Monitoring
- Vulnerability Management
- Malware Analysis
- Phishing Investigation
- Endpoint Security
- Incident Response
- SIEM Dashboard
- Security Policy Development
- System Hardening
- Final Security Assessment

Tools Covered

Operating Systems

- Windows 11
- Kali Linux

Virtualization

- VirtualBox
- VMware Workstation

Network & Monitoring

- Wireshark
- netstat
- ping
- tracert

Firewall & Endpoint Protection

- Windows Defender Firewall
- UFW
- Microsoft Defender Antivirus
- BitLocker

Vulnerability Management

- Nessus Essentials
- OpenVAS
- Microsoft Defender Vulnerability Management

Log Analysis & SIEM

- Windows Event Viewer
- Syslog
- Splunk Free
- Wazuh
- Elastic (ELK Stack)

Malware & Email Analysis

- VirusTotal
- Hybrid Analysis
- Any.Run (Community Edition)
- MXToolbox
- Google Safe Browsing

Incident Response

- TheHive
- Cortex

Security Standards

- CIS Benchmarks
- ISO/IEC 27001
- Microsoft Security Compliance Toolkit

Documentation

- Microsoft Word
- Microsoft Excel
- Microsoft PowerPoint
- GitHub



+91 63691 70669
+91 77952 03926
+91 88704 84604
+91 99452 03749



Email : admin@skystenainfotech.com ,
skystenainfotechcbe@gmail.com
Website : www.skystenainfotech.in



No.399, Shree Complex, 2nd Floor,
100 Feet Road, 6th Street Extension,
Gandhipuram, Coimbatore - 641 012.
Tamilnadu. INDIA.